



**British
Algorithmic
Olympiad**

SYLLABUS

DETAILS

This document contains the syllabus for the British Algorithmic Olympiad. This has been updated for the **November 2025** series, and contains new additions.



SYLLABUS

The British Algorithmic Olympiad will contain questions based off some the following topics and techniques. This list is not exhaustive.

Additions for the **November 2025** series have been marked with an asterisk (*).

1 Mathematics

- 1.1 **Cartesian Coordinate System:** Two and three-dimensional. Equations of lines (linear and non-linear).
- 1.2 **Functions:** Function notation. Composite and piecewise functions.
- 1.3 **Recurrence Relations:** Notation. Periodic sequences.
- 1.4 **Arithmetic and Geometric Sequences:** Sigma notation. Sums of sequences.
- 1.5 **Proof:** Contradiction, deduction, counterexample, exhaustion.
- 1.6 **Combinatorics:** Counting. Factorials.
- 1.7 **Sets:** Notation. Special sets of numbers (e.g. $\mathbb{Z}^+$). Subsets. Inclusion-exclusion.
- 1.8 **Radians*:** Use of radians and degrees. Use of radians in programming.

2 Computer Science

- 2.1 **Logic:** Understanding of sequence, selection and iteration in programming.
- 2.2 **Programming:** Familiarity with a language. Use of a range of data types. Functions, inter-function communication.
- 2.3 **Data Structures:** A wide range of data structures.
- 2.4 **Advanced Programming Techniques & Searching Algorithms:** Specialised and recursive algorithms, for example, traversal algorithms or dynamic programming. Ability to recognise problems which require such algorithms.
- 2.5 **Time Complexity:** Understanding how input constraints can affect required solution efficiency. Time complexity analysis of a simple program.
- 2.6 **Two and Three-dimensional Representations:** Using data structures to represent, for example, grids or boxes.
- 2.7 **Number Systems:** Binary. Number systems with different bases to 10.
- 2.8 **Basic File Handling*:** Reading lines from an input file.